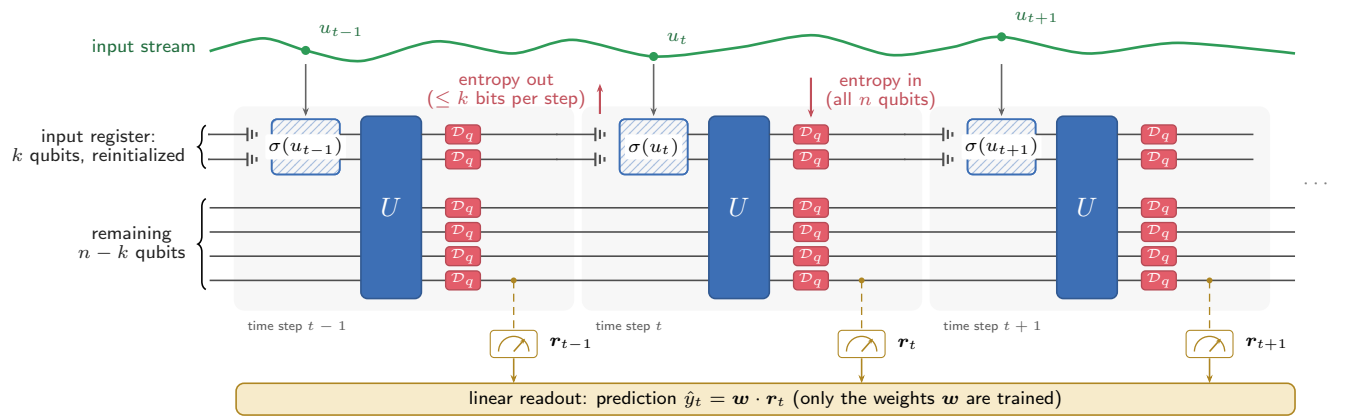

input stream
u_{t-1}
u_t
u_{t+1}
entropy out
($\leq k$ bits per step)
entropy in
(all n qubits)
input register:
k qubits, reinitialized
remaining
$n-k$ qubits
$\sigma(u_{t-1})$
$\sigma(u_t)$
$\sigma(u_{t+1})$
U
$\mathcal{D}_q$
time step $t-1$
time step t
time step $t+1$
$\boldsymbol{r}_{t-1}$
$\boldsymbol{r}_t$
$\boldsymbol{r}_{t+1}$
linear readout: prediction $\hat{y}_t = \boldsymbol{w} \cdot \boldsymbol{r}_t$ (only the weights $\boldsymbol{w}$ are trained)